ABHINAV

SOFTWARE ENGINEERING • AI/ML • DATA & PYTHON

CONTACT
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GitHub | LinkedIn

PROFESSIONAL SUMMARY

Artificial Intelligence & Machine Learning engineering graduate from RV College of Engineering with a CGPA of 8.24/10. Hands-on experience developing Python-based AI/ML workflows, LLM-powered retrieval systems, data-processing pipelines, and time-series anomaly-detection models. Additional experience in network monitoring, troubleshooting, scripting, and technical collaboration. Interested in software engineering, AI/ML engineering, Python development, and data-focused roles.

TECHNICAL SKILLS

Programming	Python, SQL basics, shell scripting
AI / Machine Learning	scikit-learn, TensorFlow/Keras, Isolation Forest, LSTM autoencoders, feature engineering, FFT-based analysis
LLM / NLP	Ollama, Qwen2.5 3B, Sentence Transformers, FAISS, BM25, hybrid retrieval, Reciprocal Rank Fusion, Cross-Encoder reranking
Data & Visualization	NumPy, Pandas, Matplotlib, Streamlit, data preprocessing, evaluation metrics
Tools & Engineering	Git/GitHub, Linux, SQLite, pytest, debugging, testing, API integration, monitoring, documentation, Agile collaboration

PROJECTS

BioRAG — Biomedical Research Assistant | **Python · FAISS · BM25 · Ollama · Sentence Transformers**

- Developed a local retrieval-augmented generation (RAG) pipeline for biomedical PDFs, combining dense vector search with BM25 keyword retrieval.
- Implemented Reciprocal Rank Fusion to combine retrieval results and Cross-Encoder reranking to prioritize relevant passages before generation.
- Integrated local Qwen2.5 3B inference and source/page-grounded responses, including a fallback when supporting evidence is unavailable.
- Added evaluation scripts to examine response latency, source coverage, Recall@5, and reciprocal-rank metrics.

GitHub: github.com/Abhinav-ai01/BioRAG

Fault Detection System | **Python · NumPy · Pandas · FFT · scikit-learn · TensorFlow/Keras**

- Built sliding-window signal analysis to process time-series data and extract statistical features such as mean, standard deviation, and frequency-domain characteristics.
- Applied FFT-based feature extraction and feature scaling to prepare signal windows for machine-learning models.
- Used Isolation Forest and an LSTM autoencoder to explore anomaly scores and reconstruction errors for potential fault identification.

GitHub: github.com/Abhinav-ai01/Fault-detection-system

JobAgentMAS — Multi-Agent Job Application Workflow | **Python · LangGraph · SQLite · pytest**

- Designed a modular workflow for job-description analysis, resume tailoring, quality review, and validation, with SQLite-backed persistence.
- Added a review/approval step to keep a human involved in the workflow and validation logic to check generated outputs.
- Used automated tests to exercise workflow behavior, demonstrating attention to modular design, testability, and maintainability.

GitHub: github.com/Abhinav-ai01/JobAgentMAS

PROFESSIONAL EXPERIENCE

AI Intern — Languify / Coincent | **Sep 2024 – Jan 2025**

- Contributed to Python-based NLP/AI workflows, including dataset preparation, preprocessing, testing, and debugging.
- Worked through iterative development and validation in Agile cycles with attention to reusable code and pipeline reliability.
- **Reported outcome:** Reduced dataset preparation overhead by 30%.

Network Administrator Intern — Avant TechnoInfra Pvt. Ltd. | **Jan 2026 – Jul 2026**

- Monitored infrastructure across 15 subnets and supported network troubleshooting, log review, and routine administration.
- Used scripting/automation to reduce repetitive manual work and assist operational monitoring.
- **Reported outcomes:** 99.5% operational uptime and 40% fewer manual-intervention/security-audit flags.

EDUCATION & CERTIFICATIONS

B.E. in Artificial Intelligence & Machine Learning — RV College of Engineering, Bengaluru | 2022–2026 | CGPA: 8.24/10

Microsoft Certified: Azure AI Fundamentals (AI-900) • Microsoft Applied Skills: Cloud & Full Stack • Microsoft Applied Skills: Data Visualization & R

LEADERSHIP & ACTIVITIES

Subsystem Layout Engineer — Team Vyoma: Supported subsystem integration and coordination across aerial test campaigns. **Core Team — E-Cell RVCE / Team DHI:** Contributed to technical events involving 500+ participants.